Isabelle McEntee

Berlin, DE | Australian citizen | linkedin.com/in/isabellemcentee

Education

Free University of Berlin — Berlin, DE
Master of Science in Physics — 2026

- **Thesis:** *Correlations in Non-Markovian Open Quantum System Dynamics* (Prof. Christiane Koch). High-dimensional stochastic simulation with long-range temporal dependence; benchmarked two numerical methods (tensor-network chain mapping, mode sampling) and characterized the accuracy vs. compute trade-off.
- **Coursework:** Advanced Quantum Mechanics, Statistical Physics & Thermodynamics, Advanced Statistical Physics, Advanced Quantum Information Theory.

Monash University — Melbourne, AU
Bachelor of Science (First Class Honours) in Physics — 2022

- **Thesis:** *Slowly Decaying Correlation in a Simple Metastable Quantum System* (Prof. Kavan Modi).
- **Coursework (master's-level):** Quantum Information Theory, Data Analysis & Machine Learning, Quantum Mechanics, Methods of Applied Mathematics, Condensed Matter Physics.

Monash University — Melbourne, AU
Bachelor of Science (Physics & Mathematics) + Bachelor of Computer Science (Data Science) — 2022

- **Coursework:** Real Analysis, Linear Algebra, Algorithms & Data Structures, Modelling for Data Analysis, Deep Learning (TensorFlow, PyTorch), Advanced Differential Equations.

Experience

Student Research Assistant — May 2024 – Present
Free University of Berlin — Berlin, DE

- Implemented and compared two methods for simulating quantum systems with long-memory (non-Markovian) environments: chain mapping with TEBD (matrix product states, orthogonal polynomials) and surrogate-Hamiltonian mode sampling.
- Showed that the valid simulation time grows linearly with chain length, independent of coupling strength, giving a rule for choosing chain length for a target time; mapped the trade-off between the methods (chain mapping reliable across regimes at ∼50 modes; surrogate Hamiltonian similar time window with 15 modes under strict sampling conditions). Presented at the DPG annual conference.

Working Student — Defect Manager (Mbition) — Mar 2024 – Apr 2026
Mercedes-Benz AG — Berlin, DE

- Investigated and resolved critical software defects via root-cause analysis on production data pipelines.
- Developed a systematic triage process using system logs and databases to classify and prioritize customer-reported issues, reducing developer response time by 50%.

Research Assistant — Jan 2021 – Jan 2023
Monash University — Melbourne, AU

- Constructed a three-body partial swap collision model and performed spectral decomposition and multi-time quantum correlation analysis, identifying anomalously slow (power-law) decay.

Software Engineer — Automation (Internship) — Jan – Jun 2020
National Australia Bank — Melbourne, AU

- Built an interface to analyse and display real-time performance metrics from AWS time-series data, pre-emptively flagging potential service disruptions before impact on banking operations.
- Contributed to automation workflows in Pega, translating business requirements into technical specifications.

Selected Projects

IMC Prosperity 4 (algorithmic trading competition) — 2026
Solo competitor — top 5% result

- Built Python strategies across all 5 rounds; strongest result in round 4, pricing options on a simulated GBM underlying and running counterparty-aware market making across a ladder of European call strikes. Other rounds: inventory-skewed market making and OU mean-reversion on cointegrated spreads.

Technical Skills

Languages: Python, Fortran, Mathematica, MATLAB, SQL, R, Java

Mathematics & Statistics: Numerical linear algebra, tensor networks, stochastic processes & SDEs, spectral methods, time-series analysis

Libraries: NumPy, SciPy, pandas, statsmodels, scikit-learn, PyTorch, TensorFlow, Matplotlib, TeNPy, QuTiP, Qiskit

Tools: Git, AWS, Linux